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Manuscript Submitted - TIP-34966-2025, A Source-Free Approach for Domain Adaptation via Multiview Image Transformation and Latent Space Consistency

1 message

Transactions on Image Processing <onbehalf@manuscriptcentral.com>

Sat, Apr 19, 2025 at 4:05 PM

Reply-To: m.langdon@ieee.org

To: sami.azam@cdu.edu.au

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19-Apr-2025

Dear Dr. Sami Azam,

Your manuscript, TIP-34966-2025, entitled "A Source-Free Approach for Domain Adaptation via Multiview Image Transformation and Latent Space Consistency" has been successfully submitted online and is presently being given full consideration for publication in the IEEE Transactions on Image Processing as a Regular Paper.

Please mention the above manuscript ID in all future correspondence or when contacting the IEEE Signal Processing Society for questions. If there are any changes in your account information or e-mail address, please log in to ScholarOne Manuscript at <https://mc.manuscriptcentral.com/tip-ieee> and edit your user information as appropriate.

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The Transactions on Image Processing has dedicated itself to providing the best possible publication opportunity. To that end, all individuals connected with the review and publication process have been requested to do their utmost to maintain the quality of the journal. We appreciate you selecting the Transactions on Image Processing.

All authors should review the below questions and the answers that the submission author provided. If you see any inaccurate information, please contact the journal administrator: Mikaela Langdon; m.langdon@ieee.org.

1) If the manuscript has more than one author, briefly describe every author's significant intellectual contribution (max 25 words per author). If this does not apply, type N/A.

2) Is this manuscript a resubmission of, or related to, a previously rejected manuscript?

No

3) If "Yes", specify the publication venue and manuscript ID of the previous submission and upload a supporting document detailing how the resubmission has addressed the concerns raised during the previous review. If this does not apply, type N/A.

4) Is this manuscript an extended version of a conference publication?

No

5) If "Yes", provide the full citation of the conference submission or publication. If this does not apply, type N/A.

6) Is this manuscript related to any other papers of the authors that are either published, accepted for publication, or currently under review, and that are not included among the references cited in the manuscript?

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8) What is the contribution of this paper, within the scope of Transactions on Image Processing?

This research directly contributes to the *IEEE Transactions on Image Processing*'s focus on advancing algorithms and architectures for image analysis, processing, and multidimensional applications. By providing a source-free framework that simplifies the domain adaptation process while maintaining robust performance, our work has implications for diverse applications, including image classification, video analysis, and biomedical imaging.

9) Why is the contribution significant (What impact will it have)?

10) What are the three papers in the published literature most closely related to this paper? Please provide full citation details, including DOI references where possible.

[1] "Minimum class confusion for versatile domain adaptation," in Computer Vision—ECCV 2020 [https://doi.org/10.1007/978-3-030-58589-1_28](https://doi.org/10.1007/978-3-030-58589-1_28) [2] "Domain adaptation without source data," IEEE Transactions on Artificial Intelligence, <https://doi.org/10.1109/TAI.2021.3110179> [3] "Source-free domain adaptation via distribution estimation," in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022 <https://doi.org/10.1109/CVPR52688.2022.00707>

11) What is distinctive/new about the current paper relative to these previously published works?

Our paper uniquely integrates multiview augmentations, source-free operation, and computational simplicity. Unlike methods relying on pseudo-labeling, pre-trained models, or adversarial training, which are costly or fail under domain shifts, our approach uses multiview augmentation and latent space consistency to learn domain-invariant features directly from the target domain, reducing costs and enhancing scalability for scenarios with restricted source data.

Sincerely,

Mikaela Langdon
TIP Staff Administrator
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